

Check Your Showerheads!

Most showerheads today use about 2.5 gallons per minute. We could switch to very low-flow showerheads that save lots of water and energy and still provide great showers.

by Jon Harrod and Lexie Hain

Replacing showerheads manufactured before 1994 saved water and energy. But for much greater savings, it may be time to replace the newer showerheads as well.

Nationally, showering accounts for about 17% of residential water use; in many households it is the largest end use of hot water. So reducing the amount of water used for showering can offer substantial savings for occupants and building owners. Reducing shower flows can also help conserve water supplies and reduce the demand on wastewater treatment facilities. But the widespread change-out to more-efficient showerheads will be successful only if people actually use them. And people will use them only if they give a good shower.

Federal legislation implemented in 1994 set the maximum flow rate for new showerheads at 2.5 gallons per minute (gpm), at 80 lb per square inch (psi) water pressure. Many showerheads manufactured prior to 1994 have flow rates of 3–7 gpm at typical residential water pressures. The 1994 regulations, together with utility-sponsored programs, have led to the widespread replacement of older showerheads. According to recent research done by Peter Biermayer at Lawrence Berkeley National Laboratory, approximately 80% of showerheads currently in use have

flows of 2.5 gpm or less. Actual flow rates are often lower than rated flows, reflecting variations in water pressure and occupants who “throttle back” shower control valves to achieve desired temperatures and/or flows. Overall, flow rates measured in the field for the Biermayer research averaged around 2.2 gpm.

Replacing any high-flow, pre-1994 showerheads that remain with modern 2.5 gpm showerheads will be highly cost-effective for individual households. However, efficiency programs that specify 2.5 gpm as the standard for new showerheads may show only modest savings. A recent article (“Saving Water Indoors,” *HE* 2007 Special Issue: Water/Energy, p. 27) reports that savings from 2.5 gpm showerhead retrofits in Seattle, Washington, and Oakland, California, were small (9% in both cases) and not statistically significant, suggesting that many showerheads already have flows of 2.5 gpm or less. In contrast, the article reported that a retrofit program in Tampa, Florida, that used 1.75 gpm showerheads produced a statistically significant 28% reduction in water used for showering.

Several showerheads are currently available with rated flows of less than 2 gpm at 80 psi water pressure. These very low-flow (VLF) showerheads offer the potential for substantial savings relative to standard 2.5 gpm



JOHN HARROD

The 1.75 gpm Niagara Chrome Earth received high marks despite its low price.

models. While VLF showerheads have been available for years, some auditors and program managers have been reluctant to recommend them. Some of this resistance may stem from experience with an earlier generation of water-saving showerheads that provided substandard showers. But like low-flush toilets, VLF showerheads have evolved over the last decade, and some are now marketed toward high-end consumers and facilities. Resistance may also reflect the perception that VLF showerheads do not provide worthwhile savings relative to 2.5 gpm models, a perception that we examine in this article.

Showering on Behalf of Science

To see for ourselves how VLF showerheads perform, we purchased seven showerheads with rated flows between 1 and 1.75 gpm, testing actual flows at city of Ithaca water pressure (70 psi with the shower running) and at 40 psi on a rural well system. We collected feedback on the showerheads from friends and family members, asking participants to rate each showerhead from poor to excellent on a scale of 1 to 5 (see Table 1). The

Table 1. Very Low Flow (VLF) Showerheads

Showerhead	Rated flow @ 80 psi	Measured flow @ 75 psi	Measured flow @ 40 psi	Purchase price	Materials*	Average score (1-5, 5= excellent)
Niagara Chrome Earth	1.75	1.9	1.0	\$ 2.95**	P/B	4.3
Whedon USB4C	1.5	1.3	1.2	\$ 8.99	B	3.3
Oxygenics 630-XLF	1.5	1.5	1.0	\$20.45	P/B	3.5
Jet-Stream E-222	1.5	1.3	1.0	\$29.95	B	2.3
Delta RP46384	1.6	1.6	1.5	\$34.19	P/B	2.8
Bricor B100MAX***	1	1.2	0.8	\$54.95	B	3.5
Bricor B150CH-R***	1.5	1.7	1.1	\$59.95	P/B	4.6

* P=plastic B=brass

** Price for Niagara Chrome Earth showerhead is wholesale.

*** Bricor showerheads are made to order for user-specified pressures. Models tested were made for 75 psi.

average ratings presented in Table 1 reflect 3 to 6 ratings per showerhead.

We purchased showerheads online from retail outlets or direct from manufacturers. Flow rates measured at city 70 psi were within 0.2 gpm of manufacturer's 80 psi ratings for all showerheads; flows at 40 psi were lower and more variable. Flow control technology was of three types: aerating showerheads, which increase perceived flow and rinsing ability by mixing air with the water stream; laminar-flow showerheads, which produce fine parallel streams; and Delta's H2Okinetic technology, which uses an engineered orifice to restrict flow while producing large droplets and the sensation of denser coverage within the spray area.

Niagara Chrome Earth. Despite its low cost, the Niagara Chrome Earth got high marks from our testers. Its pattern and force can be adjusted to meet individual preferences. One setting offers a broad fan of gentle streamlets; the other provides a narrower, pulsating massage. Purchased wholesale from the Energy Federation, Incorporated (EFI), a leading national distributor of residential energy efficiency products, this laminar-flow showerhead cost less than \$3, making it an attractive option for programs with tight budgets.

Whedon USB4C. The Whedon USB4C is a small aerating showerhead with a push-button shutoff valve that is reminiscent of many early low-flow

showerheads. It produces a forceful but relatively narrow cone of water droplets. The showerhead's short length (about 2.5 inches) may make it a good choice for installations where the shower arm is low on the wall. One of our testers found that it worked particularly well with her low-pressure rural water supply.

Oxygenics 630-XLF. Jon Harrod has used an Oxygenics 2.0 gpm aerating showerhead for years and has recommended it to several customers. Based on their feedback, we have concluded that it performs well at rural water pressures of 30–50 psi. It also seems to produce fewer fine water droplets than many other showerheads, reducing moisture accumulation from suspended droplets on mirrors and other bathroom surfaces. The look and feel of the 1.5 gpm 630-XLF model is very similar, except that this model has no shutoff valve. The showerhead produces a narrow jet that may be invigorating to some users but too forceful for others.

Jet-Stream E-222. Based on a design patented in 1978, the Jet-Stream produces a moderately intense cone of fine, aerated droplets. The manufacturer's Web site (www.jet-streamshowerhead.com) claims a 96% acceptance rate and includes scans of several testimonial letters. Like the Whedon model, the Jet-Stream is

suitable for applications requiring a shorter showerhead.

Delta RP46384.

Delta Faucet, the largest manufacturer of faucets and accessories in the United States, offers a 1.6 gpm showerhead that utilizes its H2Okinetic technology. The

flow from this largish, bell-shaped showerhead emerges from four rectangular ports. The Delta provides a moderately forceful, drenching shower with good rinsing action.

Bricor B100MAX. The Bricor B100MAX has the lowest flow rates of any showerhead in the study. It has an adjustment valve that allows the user to choose between ultrafine high-velocity streams and a slower, mellower cascade. Its low flow rate makes it a good choice for situations where saving water is the most important consideration.

Unlike other manufacturers, which sell a single model for a range of water pressures, Bricor makes its showerheads to order for user-specified pressures. According to Bricor, this allows precise balancing of flows across the floors of a high-rise apartment or hotel.

Bricor B150CH-R. At just under \$60, the Bricor B150CH-R is the most expensive showerhead that we tested; it also received the highest overall ratings. The 36 orifices on the face of the showerhead provide firm, highly aerated streams with good rinsing ability. Several testers also commented on the attractive appearance and finish of this model.

Economic Analysis

We calculated simple payback for a showerhead costing \$30 (the median cost of VLF showerheads in our study),



JOHN HARROD

The 1 gpm Bricor B100MAX had the lowest flow rate of the showerheads we tested.

at a range of costs for gas and electricity, and of reductions in gpm flow (see Table 2). Behavioral and demographic assumptions are based on those used by Biermayer: 0.7 showers per person per day, an average shower duration of 8.2 minutes, and 2.59 persons per household. To calculate the energy used to heat water, we assumed a temperature rise of 50°F and typical recovery efficiencies for water-heating equipment (0.8 for gas; 0.98 for electric).

Even for cheap fuel rates and small reductions in gpm flow, the \$30 VLF retrofit offers attractive paybacks. Replacing a 2.2 gpm showerhead (a typical flow for a showerhead meeting 1994 standards according to the Biermayer study) with a 1.5 gpm VLF showerhead will, in most cases, pay for itself in less than a year and a half. Therefore, our analysis indicates that it will be cost-effective to replace even many showerheads rated at 2.5 gpm with VLF models.

In addition to reducing water-heating energy, VLF showerheads can

Table 2. Simple Payback (in years) for a \$30 Retrofit of an Existing 2.2 gpm Showerhead

a. Natural gas water heater; recovery efficiency = 80%				b. Electric water heater; recovery efficiency = 98%			
	New showerhead flow (gpm)				New showerhead flow (gpm)		
Cost per therm	1.9	1.7	1.5	Cost per kWh	1.9	1.7	1.5
\$1.00	3.7	2.2	1.6	\$0.05	3.1	1.8	1.3
\$1.50	2.5	1.3	1.1	\$0.10	1.5	0.9	0.7
\$1.60	1.8	1.1	0.8	\$0.15	1.0	0.6	0.4

reduce municipal water and sewer charges and demand on rural well and septic systems; these benefits are not reflected in this analysis. In addition, VLF showerheads may represent a cheap fix to extend the capacity of an undersized or overtaxed water heater.

Choice and Customer Acceptance

Like any efficiency measure, VLF showerheads will be successful only if customers accept them. The Tampa showerhead retrofit program reports high levels of acceptance for 1.75 gpm showerheads: “73% of participants liked their new showerhead better than their old one.” Unlike low-flow toilets, whose performance can be quantified through standardized protocols (“The Real High-Efficiency Toilets Have Arrived,” *HE* 2007 Special Issue: Water/Energy, p. 10), satisfaction with showerheads is subjective and may reflect, among other things, skin sensitivity, hair length, water pressure, and aesthetic concerns. Each of our testers expressed enthusiasm for at least one of the VLF showerheads he or she tried; other models were met with ambivalence and, in a few cases, strong dislike. With the range of VLF showerheads now available, auditors and home performance contractors can present different options and allow customers to make the choice. If high levels of customer acceptance and satisfaction can be achieved, the widespread adoption of VLF showerheads offers a cost-effective opportunity to reduce water and energy use.

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Jon Harrod is owner and Lexie Hain is multifamily project manager at Snug Planet, a building performance contracting firm based in Ithaca, New York.

For more information:

Contact the authors at jharrod@snugplanet.com.

For Biermayer, P. J., Potential Water and Energy Savings from Showerheads, Lawrence Berkeley National Laboratory white paper, go to www.library.lbl.gov/docs/LBNL/586/01/PDF/LBNL-58601.pdf.

Niagara Chrome Earth showerheads are available for order online at www.efi.org.

Whedon USB4C showerheads are available at www.hardwarestore.com.

Oxygenics 630-XLF showerheads are available at www.efi.org.

Jet-Stream E-222 showerheads are available at www.jet-streamshowerhead.com.

Delta RP46384 showerheads are available at www.faucetdepot.com.

Bricor B100MAX and Bricor B150CH-R showerheads are available at www.bricor.com.